Towards Early Prediction of Human iPSC Reprogramming Success
Supplementary

1. Datasets

Numerical details of the annotated data generated by the first and last stages of the annotation process are given in Tables 1 and 2 respectively.

Table 1: Numerical details of the selective uncategorized labeled data generated by the first stage of the annotation process.

Frame Range	**Sequences**	**Frames**	**Cells**
146– 200	3	130	948
201 - 250	8	377	2952
251 - 275	6	149	868
146 - 275	17	656	4768

Table 2: Numerical details of the categorized retrospective labeled data generated by the last stage of the annotation process.

	Train			**Test**			**All**		
Frames	**Start**	**End**	**Total**	**Start**	**End**	**Total**	**Start**	**End**	**Total**
Early Train	163	202	1178	146	162	496	146	275	3937
Late Train	203	275	2263						
Cells	**iPSC**	**Diff**	**Total**	**iPSC**	**Diff**	**Total**	**iPSC**	**Diff**	**Total**
Early Train	1879	5380	7259	790	2467	3257	6056	16542	22598
Late Train	3387	8695	12082						

2. XGBoost Features

XGBoost classifier was trained with the following features:

- area: number of pixels enclosed by the cell
- pixel-min: maximum of the pixel values enclosed by the cell
- pixel-max: minimum of the pixel values enclosed by the cell

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- pixel-std: standard deviation of the pixel values enclosed by the cell
- circlicity: ratio of the number of pixels in the circumcircle of the cell to that in the cell itself.
- speed: $\sqrt{dx^2 + dy^2}$ where (dx, dy) is the change in location of the cell centroid from the previous frame
- displacement: $\sqrt{init_dx^2 + init_dy^2}$ where $(init_dx, init_dy)$ is the change in the location of the cell centroid from the first frame

3. Tracking Algorithm

A simplified high-level algorithm describing the interactive semi-automated tracking process used for retrospective labelling is given in Alg. 1.

Algorithm 1 Human-Assisted Tracking for Retrospective Labeling

```
prev_tracks ← ∅
for frame in ROI sequence from last to first do
    curr_dets ← cells detected in current frame by the instance segmentation model
    prev_tracks ← tracked cells from previous frame if any else ∅
    curr_tracks ← ∅
    for each pair of cells in curr_dets with IOU > 0.5 do
        Ask user to choose the duplicate cell from the pair if any and discard all such cells
    end for
    Perform IOU-based pairwise association between prev_tracks and curr_dets

    for each unassociated cell in curr_dets do
        Ask user to decide if it is a false positive and discard it if so
    end for
    for each unassociated track in prev_tracks do
        Ask user to associate it with a cell in curr_dets or discard it as having ended
    end for
    for each pair of associated cells with a relative change in height, width or area > 0.5 do
        Ask user to choose if this ambiguous association represents cell division, fusion or misassociation and update
    tracks accordingly
    end for
    Extend each associated track in prev_tracks with the corresponding cell from curr_dets and add the extended
    track to curr_tracks
    Add a new track to curr_tracks for each remaining unassociated cell in curr_dets
    prev_tracks ← curr_tracks
end for
```

4. Temporal Detection Metrics Results

Fig. 1 shows the impact of subsequence length on the detection metrics of RP-AUC and AP. Similar to ROC-AUC, there is a general trend of performance improvement with increase in subsequence length though not as strong as ROC-AUC and even an inverse trend being apparent in several cases.

5. Visualization Videos

All visualization videos are available in this Hugging Face repository folder:
https://huggingface.co/abhineet123/ipsc_prediction/tree/main/supplementary/visualizations.

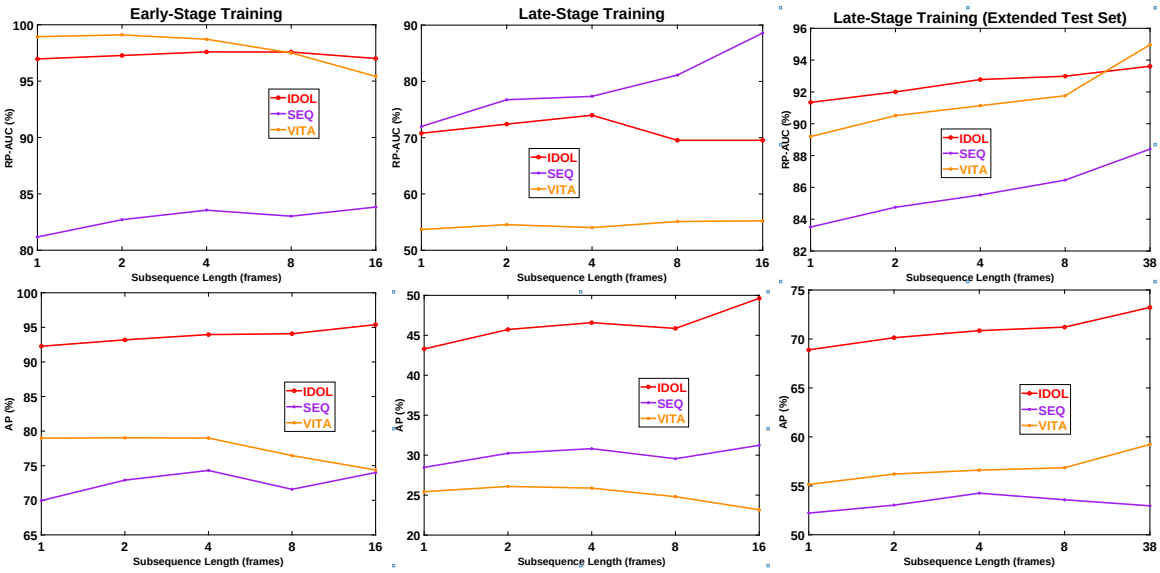

Figure 1: Subsequential results for (top) RP-AUC and (bottom) AP. Left and center plots show early and late-stage models tested on the standard test set of frames 146 - 162 while the right one shows the latter tested on an extended test set with frames 146 - 201. Note the curtailed and variable Y-axis ranges.

There are four types of videos, each in its own subfolder:

- detection failures: 2 FPS videos showing all 6 types of detection failures for each detector in both early and late-stage training configurations
  - FP videos contain one frame for each detection failure so that each source frame is present multiple times if it contains multiple FPs
  - FN videos show all the missing detections for each source frame together so each source frame is present only once
  - each frame contains three images showing, from left to right, GT annotations, all detections in the frame and a single FP detection or GT cells corresponding to all the FNs
  - in case of FP-NEX-PART, the third frame shows both the FP detection and the GT cell with which it partially overlaps
  - first two images show iPSCs and DfCs in green and red respectively while the third image shows FPs in magenta and GTs in cyan
  - black bar at the bottom of each frame shows the ROI and frame info in the first line while the second line shows some stats that can be ignored
- detection results: 2 FPS videos showing the GT and detected cells for all source frames in each of the 31 ROIs
  - iPSCs and DfCs are shown in green and red respectively
- retrospective labeling: 2 FPS videos showing the result of retrospective labelling for each of the 31 ROIs

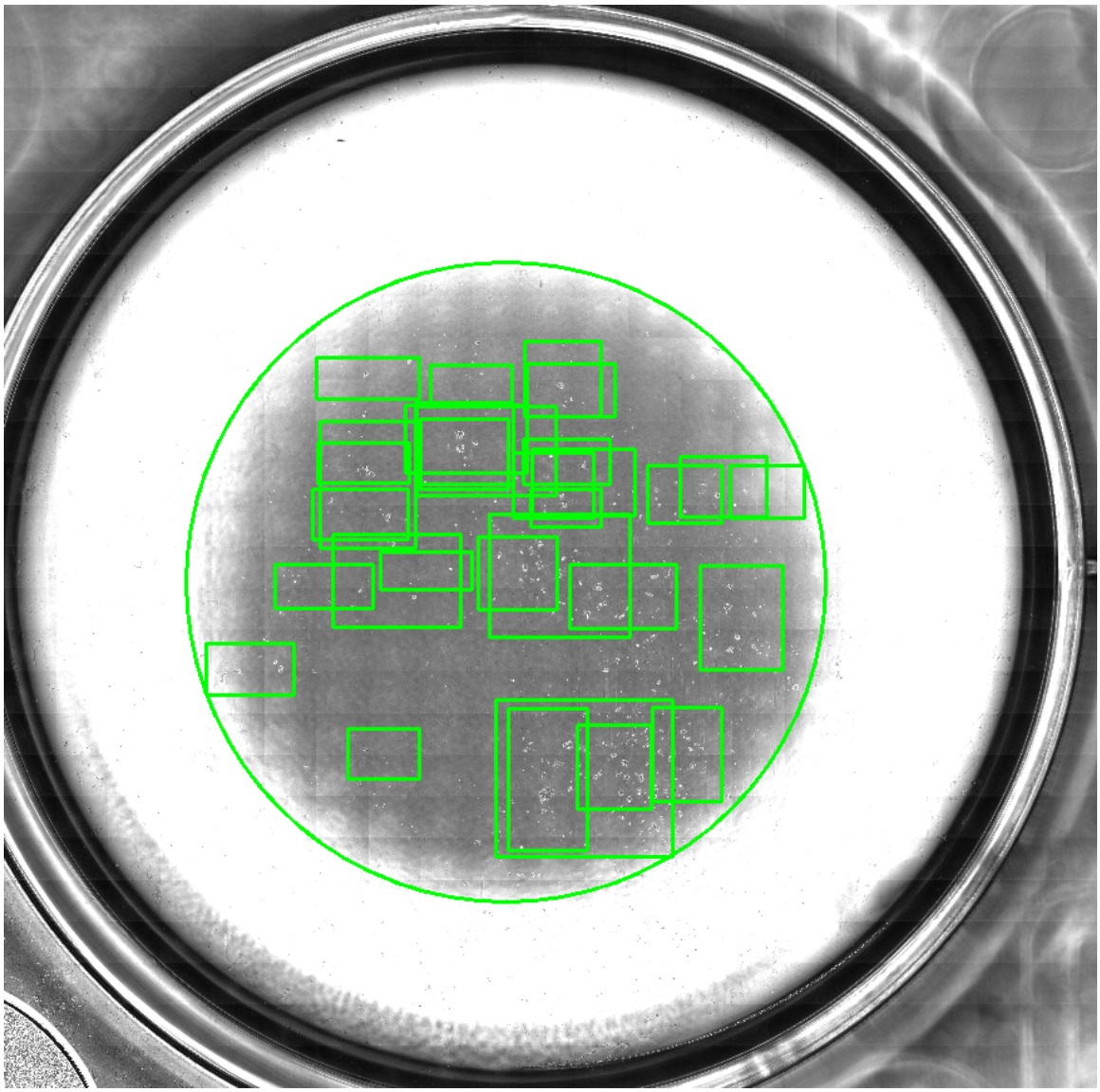

Figure 2: A sample 714 MP raw image with the 31 ROIs shown in green bounding boxes.

  - frames are ordered backwards in time since that is how the tracking was done
  - the mask colour of each cell indicates its unique identity while the bounding box colour indicates its class – green for iPSC and red for DfC
- cell lines: a single 5 FPS video showing the 8 cell lines generated by backwards tracking of a single target in one of the ROIs
  - cell lines are arranged in a 2×4 grid
  - frames are ordered backwards in time
  - a cell line is considered to have diverged into two by a cell division event while a fusion event terminates both parent cell lines and creates a new one
  - multiple cell lines have many frames in common but each has at least one frame with a unique cell-identity that makes this line distinct from others

All videos are formatted as MPEG-4 with mp4v codec and should be playable on any video player on Linux, Windows or Mac.

6. 3D Interactive Plots

Frame-wise partial AUC plots are available as interactive HTML files in this Hugging Face repository folder:
https://huggingface.co/abhineet123/ipsc_prediction/tree/main/supplementary/3d_plots.
These can be opened in any web browser.